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FILM COOLING MEASUREMENTS FOR A LAIDBACK FAN-SHAPED HOLE – EFFECT OF COOLANT CROSSFLOW ON COOLING EFFECTIVENESS AND HEAT TRANSFER

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ABSTRACT

Internal coolant passages of gas turbine vanes and blades have various orientations relative to the external hot gas flow. As a consequence, the inflow of film cooling holes varies as well. To further identify the influencing parameters of film cooling under varying inflow conditions, the present paper provides detailed experimental data. The generic study is performed in a novel test rig which enables compliance with all relevant similarity parameters including density ratio. Film cooling effectiveness as well as heat transfer of a 10-10-10deg laidback fan-shaped cooling hole are discussed. Data are processed and presented over 50 hole diameters downstream of the cooling hole exit.

First, the parallel coolant flow setup is discussed. Subsequently, it is compared to a perpendicular coolant flow setup at a moderate coolant channel Reynolds number. For the perpendicular coolant flow, asymmetric flow separation in the diffuser occurs and leads to a reduction of film cooling effectiveness. For a higher coolant channel Reynolds number and perpendicular coolant flow, asymmetry increases and cooling effectiveness is further decreased. An increase in blowing ratio does not lead to a significant increase in cooling effectiveness. For all cases investigated, heat transfer augmentation due to film cooling is observed. Heat transfer is highest in the near hole region and decreases further downstream. Results prove that coolant flow orientation has a severe impact on both parameters.

NOMENCLATURE

A	Area
D	Cooling hole diameter
DR	Density ratio ($=\rho_c/\rho_h$)
L	Cooling hole length
M	Blowing ratio ($=(\rho_c u_c)/(\rho_h u_h)$)
P	Cooling hole pitch
Re	Reynolds number
T	Temperature
Tu	Turbulence intensity
h	Heat transfer coefficient
l_ε	Turbulent length scale
p	Pressure
$\dot{q}$	Specific convective heat flux
r	Radius
u	Free stream velocity
x	Streamwise coordinate
y	Lateral coordinate
z	Wall-normal coordinate
Greek	
α	Ejection angle
δ_1	Displacement thickness
η	Film cooling effectiveness
λ	Thermal conductivity
ρ	Density

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Subscripts

CL	Centerline
aw	Adiabatic wall
c	Coolant
cc	Cooling channel
f	Film cooling
h	Hot gas
lat	Laterally averaged
rec	Recovery
t	Total
w	Wall
0	Baseline (without film cooling)

Abbreviations

CFD	Computational Fluid Dynamics
ITS	Institute of Thermal Turbomachinery
PEEK	Polyether ether ketone
TiAl	Titanium aluminide (TiAl6V4)

INTRODUCTION

High turbine inlet temperatures are imperative when a high gas turbine efficiency is to be reached. As a consequence, sophisticated cooling techniques are required to keep material temperatures within tolerable limits. This particularly applies to first stage blades and vanes of gas turbines. They are cooled extensively with both internal convective cooling and external film cooling.

Film cooling is influenced by a variety of parameters, such as blowing ratio, density ratio (Schroeder and Thole [1]), turbulence intensity (Saumweber et al. [2]), boundary layer thickness (Anderson et al. [3]), hot gas Reynolds number ([3]), cooling hole shape (Bunker [4]), and surface roughness (Schmidt et al. [5]). A comprehensive overview is given by Bogard and Thole [6]. A major aspect not considered in many studies is the influence of coolant crossflow at the hole inlets.

Vanes and blades are equipped with internal channels that are passed by a flow in order to achieve a high internal heat transfer. Film cooling holes are fed by these internal channels, i.e. in general, crossflow is present at the inlet of a cooling hole. This crossflow has a major impact on film cooling performance.

To gain a more in-depth understanding of the influencing parameters, a new film cooling test rig was built at the Institute of Thermal Turbomachinery (ITS), see Fraas et al. [7]. The test rig provides for different coolant flow conditions at the hole inlet. This study focuses on coolant flow effects on film cooling effectiveness and heat transfer for a state-of-the-art laidback fan-shaped hole.

LITERATURE REVIEW

Coolant flow at the inlet of a film cooling hole affects both film cooling effectiveness and heat transfer. In general, two flow situations can be distinguished: Coolant flow at the inlet is either parallel or perpendicular to the hot gas flow.

Under plenum conditions in the cooling channel, flow separation can be observed at inclined cooling holes, see e.g. Leylek and Zerkle [8]. Parallel coolant flow influences this flow separation. Increasing flow velocity in the coolant channel leads to a gradual reduction of the size of the separation bubble at the downstream edge of the cooling hole. At higher flow velocities, a separation bubble will form at the upstream edge of the hole inlet, see Acharya and Houston Leedom [9]. This will not only influence the losses, but also the velocity profile at the exit of the cooling hole, as is shown by Thole et al. [10]. A larger amount of air is ejected at the downstream edge of the cooling hole exit, leading to reduced mixing between coolant and hot gas and, hence, to a higher film cooling effectiveness at high blowing ratios, see Gritsch et al. [11] and Saumweber et al. [12]. This holds true for cylindrical and diffuser-shaped holes. However, Gritsch et al. [13] state that heat transfer is also increased in case of a parallel coolant flow, since turbulence is increased in the cooling jet.

Perpendicular coolant flow, on the other hand, has a more pronounced influence on film cooling performance. Perpendicular inflow induces a helical flow inside of the cooling hole. This is shown by Kohli and Thole [14] and Saumweber and Schulz [15] using CFD. The helical motion stabilizes the cooling jet such that it remains attached to the surface and leads to a wider lateral extension of the jet and a skewed cooling footprint on the surface, see Saumweber et al. [12]. Film cooling effectiveness is slightly reduced for low blowing ratios, but increased for high blowing ratios (Gritsch et al. [16]). Results of Saumweber and Schulz [15] show that heat transfer is also increased due to the intensified contact between hot gas and coolant flow. For cylindrical compound angle holes, the performance depends on inflow orientation. McClintic et al. [17] found that for an inline inflow (45° deflection), performance increased at high blowing ratios. For a counter inflow (135° deflection), on the other hand, performance increased at low blowing ratios and decreased rapidly at high blow ratios. For fan-shaped and laidback fan-shaped cooling holes, an asymmetric flow separation is induced inside the diffuser, leading to an asymmetric coolant footprint on the surface and reduced film cooling effectiveness for most cases (see Saumweber et al. [12], Gritsch et al. [16], Wilkes et al. [18], and McClintic et al. [19]). According to Saumweber and Schulz [15], heat transfer is reduced for a fan-shaped hole with perpendicular coolant flow due to a reduced contact between cooling jet and surface. The influence of perpendicular coolant flow for compound angle shaped holes is investigated by Dittmar et al. [20] and McClintic et al. [21]. Their results proof that flow direction (inline or counter inflow) has a major impact on the diffuser separation and, consequently, on the cooling performance. However, their findings differ. Dittmar et al. [20] determined a higher effectiveness for inline inflow, whereas McClintic et al. [21] found

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that effectiveness was higher for counter inflow. Hence, it is impossible to draw general conclusions for such a complex system which depends on multiple test parameters and cooling hole geometry.

Different approaches to scaling coolant flow for experiments have been chosen: Scaling using the coolant flow Mach number (Gritsch et al. [11, 16] and Saumweber et al. [12, 15]) and scaling using the coolant channel Reynolds number based on the hydraulic diameter (McClintic et. al [17]) or the cooling hole diameter (Sakai and Takahashi [22]). Wilkes et al. [18] examine the effect of coolant flow Reynolds number (based on the hydraulic diameter of the channel) and the velocity ratio between coolant channel flow and hot gas flow u_{cc}/u_h by varying these parameters independently. Their results prove that both parameters have an effect on the cooling performance. However, the influence of u_{cc}/u_h is dominant. Further research by McClintic et al. [19] shows that the discharge coefficient as well as the lateral shift of peak effectiveness scale well with the velocity ratio between the coolant channel flow and the coolant flow inside the hole u_{cc}/u_c. Fraas et al. [7] also proposed that the parameter u_{cc}/u_c is crucial to scaling the coolant channel velocity. They stated that matching of the coolant channel Reynolds number, the hot gas channel Reynolds number (both based on cooling hole diameter), density ratio, and blowing ratio leads to a realistic velocity ratio u_{cc}/u_c as well.

To conclude, several studies revealed the detrimental effect of perpendicular coolant flow for fan-shaped and laidback fan-shaped film cooling holes. However, accessible data for different geometries are still limited. Data for heat transfer augmentation are available for the near-hole area only. Additionally, heat transfer was only measured at a low turbulence level of 5 % (Saumweber and Schulz [15]) or 2 % (Gritsch et al. [13]) for a perpendicular and parallel coolant flow, respectively. Data from Saumweber et al. [2], however, reveal that at low turbulence levels, trends can even be reversed compared to a high turbulence case. Therefore, this study aims at investigating coolant flow effects for a state-of-the-art laidback fan-shaped hole at a high free stream turbulence level. Data are presented over 50 cooling hole diameters downstream of the cooling hole exit. A better insight in heat transfer augmentation is given and a detailed analysis of coolant flow velocity effects is conducted. For this, a low and a very high coolant flow Reynolds number are chosen.

EXPERIMENTAL SETUP AND DATA PROCESSING

In this section, an overview of the test rig used in this study and the investigated cooling hole is given, followed by a description of data processing. A more detailed discussion of the test rig, experimental boundary conditions, measurement setup, data processing, and the measurement uncertainty is given by Fraas et al. [7].

Experimental Setup

The test rig as shown in Fig. 1 consists of a hot gas and cooling channel. In order to achieve the desired flow conditions in terms of turbulence level and boundary layer state, a turbulence grid ① and a boundary layer bleed ② are installed in the inflow section of the hot gas channel. The conditions of the approaching hot gas flow are discussed by Fraas et al. [7].

Hot gas channel and cooling channel are connected via the ejection module ③. Five parallel cooling holes with axes aligned parallel to the hot gas channel are used to account for interactions between the cooling jets. PEEK is utilized as material for the ejection module in order to minimize heat transfer inside the cooling hole. This high-temperature-resistant plastic material has a very low thermal conductivity of $\lambda = 0.27\,\mathrm{W\,m^{-1}\,K^{-1}}$. The orientation of the cooling channel can be changed such that either a parallel or a perpendicular coolant flow orientation is obtained. The origin of the coordinate system is placed at the intersection of the central cooling hole axis and the surface. It is a right-handed trihedron, as a result of which coolant channel flow is in the negative y-direction for perpendicular channel orientation. The test plate ⑤ extends by more than 50 D downstream of the cooling hole outlet, enabling measurements over a large distance. Five infrared-transmissive sapphire windows ④ are placed in the channel top wall to ensure optical access over the entire measurement area.

Test conditions of the experiments are summarized in Table 1. Great effort has been made to enable transferability of the results to engine conditions, see Fraas et al. [7]. Note that not the velocity ratio between cooling channel flow and hot gas flow, but the Reynolds numbers are set during the experiments. Consequently, the values given for the velocity ratio vary slightly for different blowing ratios.

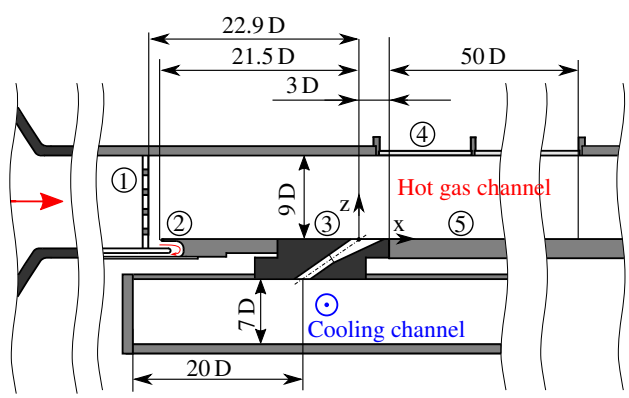

FIGURE 1: Test rig in perpendicular coolant flow configuration (adjusted from Fraas et al. [7])
1) Turbulence grid 2) Boundary layer bleed 3) Ejection module 4) Sapphire windows 5) Test plate

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TABLE 1
Experimental parameters and test conditions

Parameter	Variable	Value
Total hot gas temperature	$T_{\mathrm{t,h}}$	510 K
Total coolant temperature	$T_{\mathrm{t,c}}$	300 K
Density ratio	DR	1.7
Blowing ratio	M	0.5...3.0
Reynolds number hot gas	$Re_{\mathrm{D,h}}$	13×10^3
Reynolds number cooling channel	$Re_{\mathrm{D,cc}}$	5×10^3 ; 30×10^3
Velocity ratio cooling channel to hot gas	$u_{\mathrm{cc}}/u_{\mathrm{h}}$	$\approx$0.15 ; 0.9
Turbulence intensity	Tu_{h}	8.2 %
Turbulent length scale	l_{ε}	0.73 D
Cooling hole diameter	D	10 mm
Pitch	P	$8D$
Displacement-thickness	δ_{l}	$0.05D$

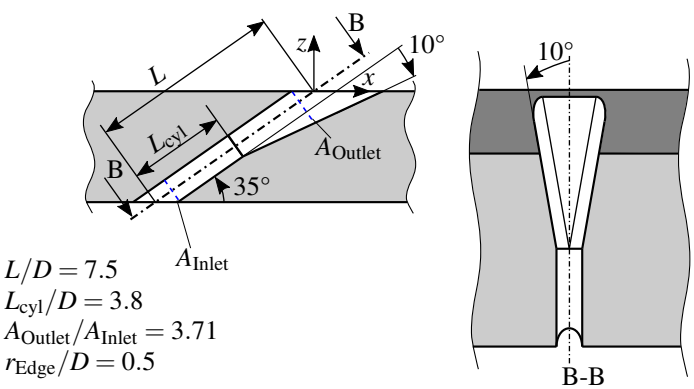

FIGURE 2: Geometry of the laidback fan-shaped hole used in this study

The geometry of the laidback fan-shaped cooling hole is given in Fig. 2. It is inclined by $\alpha = 35°$, its length is 7.5 D, and the diffuser opens with 10°. The four edges of the diffuser are rounded with 0.5 D and the inlet and breakout edges are sharp. Coverage ratio (hole breakout width based on the pitch) is 0.35. The cooling hole is in line with the main flow direction.

Measurement Setup and Data Processing

Film cooling is used to reduce the wall-normal heat flux at a given wall temperature T_{w}. According to Choe et al. [23], the heat flux can be calculated as

$$\dot{q}(x,y) = h_{\mathrm{f}}(x,y) \cdot [T_{\mathrm{aw}}(x,y) - T_{\mathrm{w}}(x,y)] \tag{1}$$

and depends on the heat transfer coefficient with film cooling h_{f} and the adiabatic wall temperature T_{aw}. Both values need to be determined experimentally. In this study, this is done by using the superposition approach, see Choe et al. [23]. This approach has already been successfully used and validated by the ITS (see Gritsch et al. [24] and Kneer et al. [25,26]).

Two measurements at different wall heat fluxes but under identical flow conditions are required for the superposition approach. Since it is beneficial for the measurement uncertainty to achieve two measurements with considerably different heat fluxes, two measurement plates are used in this study. The high heat flux test plate consists of TiAl and is cooled with water at the backside, whereas the close-to-adiabatic test plate consists of PEEK. Surface temperatures are measured using an in-situ calibrated infrared camera. For in-situ calibration, 0.25 mm type K thermocouples are placed into small notches beneath the surface of the test plate. Data processing and in-situ calibration for the raw infrared images is conducted according to Ochs et al. [27,28]. Five separate infrared images are taken and merged to cover the whole measurement area. In the overlapping area a weighted averaging algorithm is used.

Subsequently, a finite element analysis is performed for both measurement plates to determine the wall-normal heat flux. The required temperature distribution at the backside of the test plate is interpolated from discrete thermocouple readings. With the wall temperatures and the wall-normal heat flux known, the heat transfer coefficient, the adiabatic wall temperature, and the adiabatic film cooling effectiveness

$$\eta_{\mathrm{aw}} = \frac{T_{\mathrm{rec,h}} - T_{\mathrm{aw}}}{T_{\mathrm{rec,h}} - T_{\mathrm{t,c}}} \tag{2}$$

can be calculated. The adiabatic film cooling effectiveness is a function of the hot gas recovery temperature $T_{\mathrm{rec,h}}$, the adiabatic wall temperature T_{aw}, and the total coolant temperature $T_{\mathrm{t,c}}$. Heat transfer variation due to film cooling (h_{f}/h_0) is determined using the measured heat transfer without film cooling (see Fraas et al. [7]).

Measurement uncertainties are calculated with a procedure proposed by Kline and McClintock [29]. The heat transfer coefficient h_{f} can be specified with an uncertainty of less than 12 % in all areas where the adiabatic film cooling effectiveness is below $\eta_{\mathrm{aw}} = 0.7$. Uncertainty of the adiabatic film cooling effectiveness is ≤ 0.012.

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RESULTS AND DISCUSSION

Results of the film cooling measurements are discussed in this section. The operating point is set according to the parameters given in Table 1. Results are shown for a parallel (‖) as well as for a perpendicular (⊥) coolant flow orientation. Cooling channel Reynolds number $Re_{D,\mathrm{cc}}$ is varied from 5000 to 30000 for the perpendicular coolant flow direction. Blowing ratio is varied from $M = 0.5$ to $M = 3.0$.

Parallel Coolant Flow

Results for the adiabatic film cooling effectiveness along the centerline in streamwise direction are shown in Fig. 3 for $5 \leq x/D \leq 50$. Cooling channel Reynolds number is kept constant at $Re_{D,\mathrm{cc}} = 5000$ and cooling channel orientation is set to parallel coolant flow (‖). Results are shown for blowing ratios $M = 1.0$, 2.0, and 3.0. Adiabatic film cooling effectiveness decreases steadily in streamwise (x-)direction for all blowing ratios. Highest slopes occur near the cooling hole and decrease with increasing surface length. The case with the lowest blowing ratio $M = 1.0$ shows the lowest values for the adiabatic film cooling effectiveness ranging from $\eta_{\mathrm{aw}} \approx 0.6$ at $x/D = 5$ to $\eta_{\mathrm{aw}} \approx 0.12$ at $x/D = 50$. Compared to the other blowing ratios, results for $M = 3.0$ show the highest film cooling effectiveness from $\eta_{\mathrm{aw}} \approx 0.68$ at $x/D = 5$ to $\eta_{\mathrm{aw}} \approx 0.27$ at $x/D = 50$. This could be expected because of the higher amount of coolant ejected. All coolant jets seem to be attached to the measurement surface.

Figure 4 shows the results for the laterally averaged adiabatic film cooling effectiveness $\eta_{\mathrm{aw,lat}}$ for the same operating conditions. Values for $\eta_{\mathrm{aw,lat}}$ range from 0.08 to 0.28. Effectiveness for all three blowing ratios is approximately equal at $x/D = 5$, but the decrease differs as a function of the different blowing ratios. Again, the lowest blowing ratio $M = 1.0$ shows the lowest adiabatic film cooling effectiveness. Values for the intermediate blowing ratio $M = 2.0$ are generally higher than those for $M = 1.0$. Close to the cooling hole for $x/D \leq 20$, they are even higher than those for $M = 3.0$. Due to the lower amount of coolant, this is unexpected in the first instance. The highest blowing ratio $M = 3.0$ shows the highest values for the laterally averaged film cooling effectiveness at $x/D > 24$, reaching $\eta_{\mathrm{aw,lat}} \approx 0.16$ at the downstream end of the measurement plate at $x/D = 50$. The findings can be explained by the contours of the adiabatic film cooling effectiveness η_{aw} in Fig. 5a, which reflects coolant distribution close to the wall. The coolant jet is clearly visible and leads to a local decrease of the adiabatic wall temperature. Moreover, the decrease in adiabatic film cooling effectiveness with increasing surface length is clearly visible. Without exception, all of the contours for the parallel coolant flow orientation are symmetric to the centerline ($y/D = 0$). They differ quantitatively and also the spreading behavior of the coolant jets differs for the three blowing ratios. The contraction of the coolant jet is larger for the highest blowing ratio $M = 3.0$ than for the medium blowing ratio $M = 2.0$, which explains the discrepancies in the laterally averaged film cooling effectiveness discussed earlier. Such a contraction at high blowing ratios was reported by Schroeder and Thole [1]. It occurs because of the high jet momentum which results in a separation of the coolant jet at the sidewalls of the diffuser.

Results are now compared with data from Schroeder and Thole [1]. Figure 6 shows a comparison of adiabatic film cool-

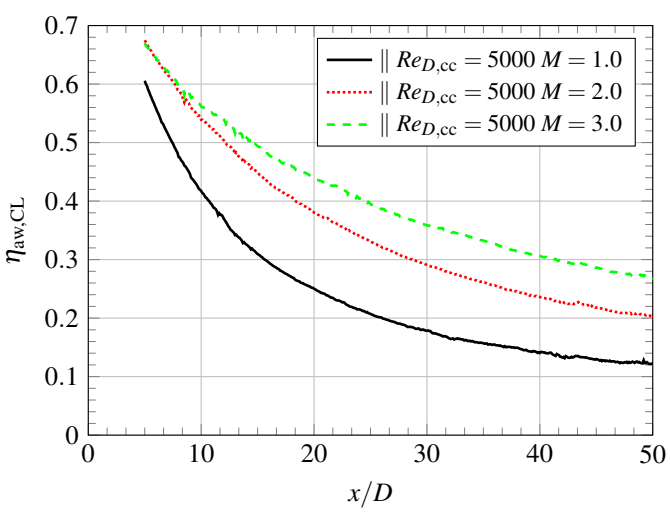

FIGURE 3: Adiabatic film cooling effectiveness η_{aw} along the centerline at $M = 1.0; 2.0; 3.0$ against $5 \leq x/D \leq 50$ (parallel coolant flow (‖), $Re_{D,\mathrm{cc}} = 5000$)

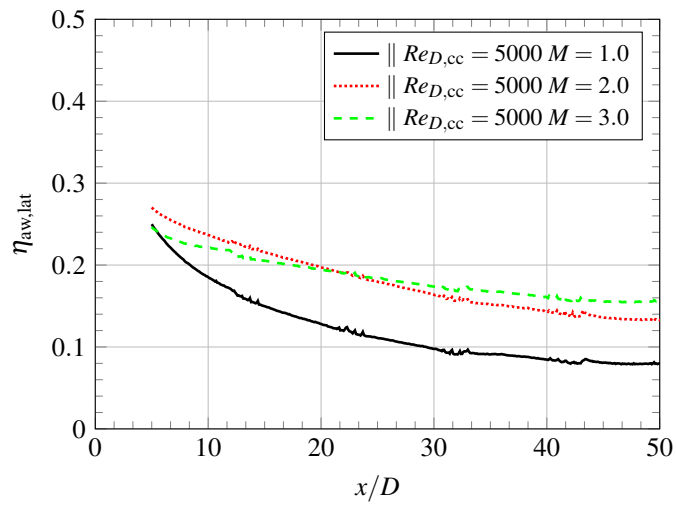

FIGURE 4: Laterally averaged adiabatic film cooling effectiveness $\eta_{\mathrm{aw,lat}}$ at $M = 1.0; 2.0; 3.0$ against $5 \leq x/D \leq 50$ (parallel coolant flow (‖), $Re_{D,\mathrm{cc}} = 5000$)

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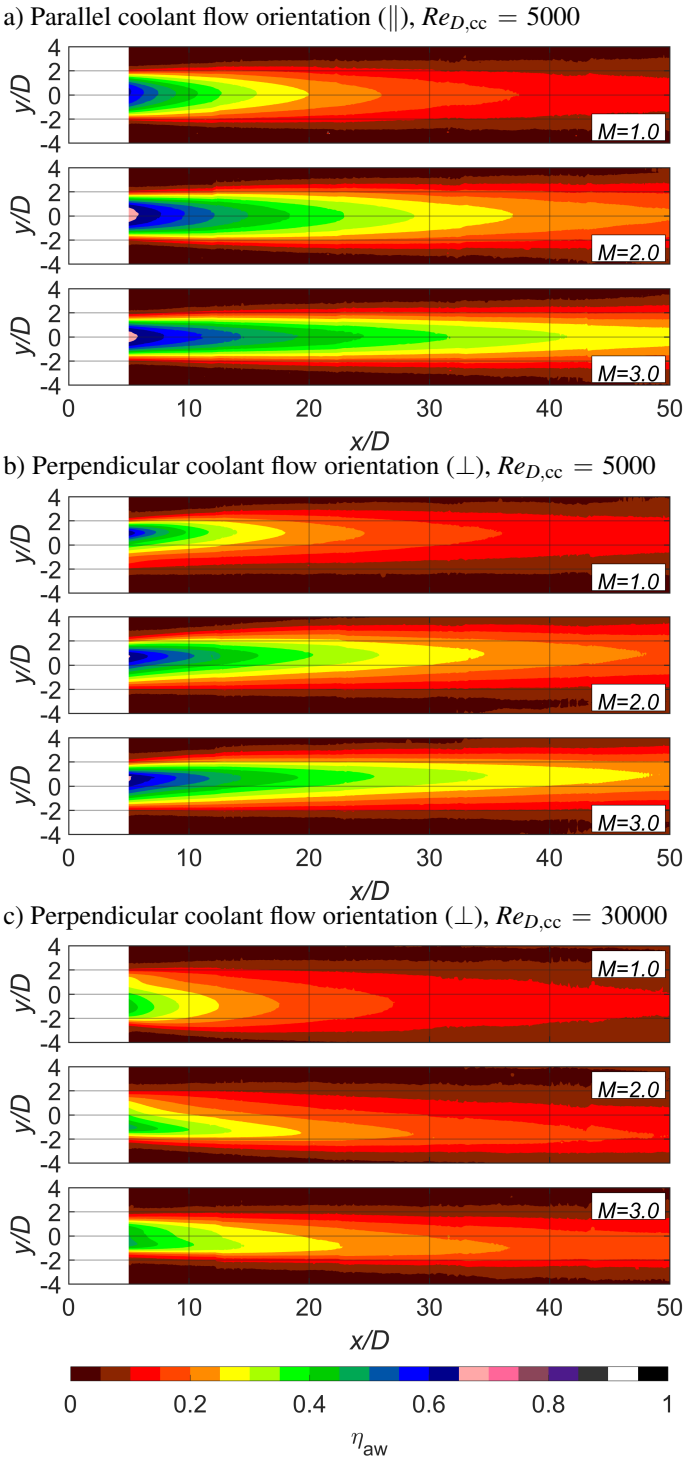

FIGURE 5: Contours of adiabatic film cooling effectiveness η_{aw} for $5 \leq x/D \leq 50$ and $-4 \leq y/D \leq 4$ at $M = 1.0; 2.0; 3.0$ and different coolant flow conditions

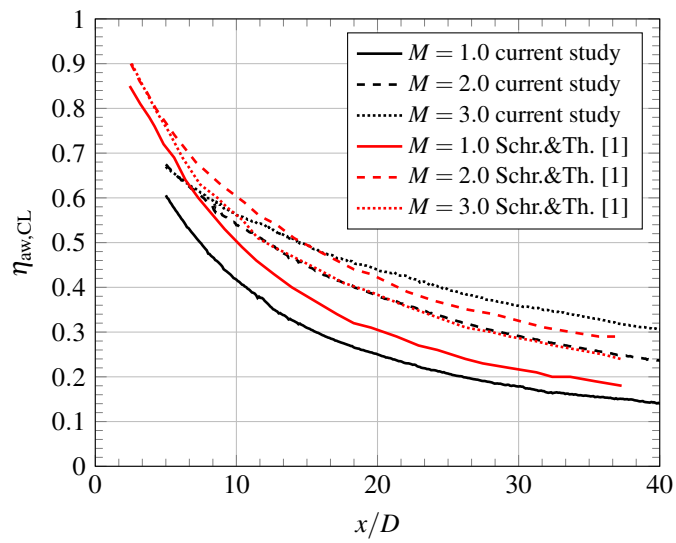

FIGURE 6: Comparison of adiabatic film cooling effectiveness η_{aw} along the centerline with data from Schroeder and Thole [1] for $M = 1.0; 2.0; 3.0$ and $5 \leq x/D \leq 40$

ing effectiveness along the centerline for the three blowing ratios $M = 1.0$, 2.0, and 3.0 and for $5 \leq x/D \leq 40$. Schroeder and Thole carried out their experiments under plenum coolant inflow conditions. Moreover, the turbulence intensity in their study was $Tu_h = 0.5\,\%$ compared to $Tu_h = 8.2\,\%$ in this study. Furthermore, the shape of the cooling hole differed from the one investigated in this study. Schroeder and Thole used a laidback fan-shaped hole with lower angles (7-7-7) compared to the 10-10-10-hole in this study. Thus, the results are not expected to match quantitatively.

Data are in excellent qualitative agreement. However, the centerline film cooling effectiveness of the current study is lower in the region close to the cooling hole exit for all blowing ratios. In the downstream region, the lines approach each other for the blowing ratios $M = 1.0$ and $M = 2.0$. The quantitative differences can be explained with the different cooling hole shapes. The lower lateral angle of the cooling hole of Schroeder and Thole leads to a lower lateral spread of the coolant and, as a consequence, adiabatic film cooling effectiveness along the centerline is higher. This effect balances out in the downstream region and can also be found in the work of Saumweber and Schulz [30]. Consequently, lines approach further downstream. It is believed that the lines would meet even further upstream if the turbulence intensity would be the same. However, the higher turbulence intensity for the current study shifts the position where the lines meet further downstream. The differences for the highest blowing ratio ($M = 3.0$) can also be attributed to the cooling hole shape. The diffuser of the cooling hole of the current study performs better for higher blow-

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ing ratios than the one used by Schroeder and Thole. This was expected because of the higher area ratio and was also investigated by Haydt et al. [31]. The different cooling hole ejection angle ($\alpha_{\text{Schroeder and Thole}} = 30° < \alpha_{\text{Current Study}} = 35°$) and length ($L/D_{\text{Schroeder and Thole}} = 6 < L/D_{\text{Current Study}} = 7.5$) are believed to be of minor relevance to the effects.

Effect of Coolant Flow Orientation

The coolant flow direction is changed in this subsection in order to investigate its effect on both adiabatic film cooling effectiveness and heat transfer. Results are shown for both parallel ($\|$) and perpendicular ($\perp$) coolant flow orientation at a constant cooling channel Reynolds number $Re_{D,\text{cc}} = 5000$. Coolant flow direction for the perpendicular flow orientation is the negative y-direction.

The effect of coolant flow orientation can be discussed qualitatively using contour plots of the adiabatic film cooling effectiveness. Figure 5b shows these contours for the blowing ratios $M = 1.0$, 2.0, and 3.0 and perpendicular coolant flow orientation. The largest difference compared to the parallel coolant flow orientation is the asymmetric footprint of the adiabatic film cooling effectiveness. This behavior (see e.g. Gritsch et al. [16]) can be explained by the diffuser aerodynamics of the shaped film cooling hole investigated. Perpendicular flow of the coolant leads to one-sided flow separation in the diffuser and, thus, to an asymmetric characteristic of the coolant jet.

Figures 7a and b show the distribution of adiabatic film cooling effectiveness for the blowing ratios $M = 1.0; 3.0$ and both cooling channel orientations at $x/D = 5$ and $x/D = 40$, respectively. This helps to further analyze the asymmetry indicated by the contour plots. At a position close to the cooling hole exit at $x/D = 5$, a deformation of adiabatic film cooling effectiveness can be observed for the perpendicular coolant flow orientation. This deformation is most pronounced for the lowest blowing ratio $M = 1.0$. Here, the maximum adiabatic film cooling effectiveness is even higher than in case of parallel orientation of the coolant flow. However, the position of the maximum value is shifted from the centerline to $y/D \approx 1$. Note that the coolant flow direction is the negative y-direction and the shift takes place towards the opposite side. Moreover, the spreading behavior differs largely and the imprint of the coolant jet on the measurement surface is much more narrow than for parallel coolant flow. For the higher blowing ratio $M = 3.0$, this is not the case. The maximum values of adiabatic film cooling effectiveness are almost the same, but the location of the maximum is also shifted towards the positive y-direction. The width of the cooled area is much larger than in case of the lower blowing ratio and the same coolant flow orientation. The position further downstream at $x/D = 40$ is shown in Fig. 7b. For this case, the maxima of the adiabatic film cooling effectiveness for both coolant flow orientations are approximately the same for $M = 1.0$ and the location of the maximum remains unchanged compared to $x/D = 5$. For $M = 3.0$, however, the position of the maximum adiabatic film cooling effectiveness has changed slightly compared to the upstream position. The location of the maximum has moved even further in the positive y-direction and the value is somewhat lower for the perpendicular coolant flow orientation compared to the parallel one. Different jet behaviors for low and high blowing ratios can be attributed to the decreasing velocity ratio between coolant channel flow and cooling hole velocity $u_{\text{cc}}/u_{\text{c}}$ (≈ 0.26 to ≈ 0.09). For low velocity ratios, the impact of perpendicular coolant flow orientation is reduced.

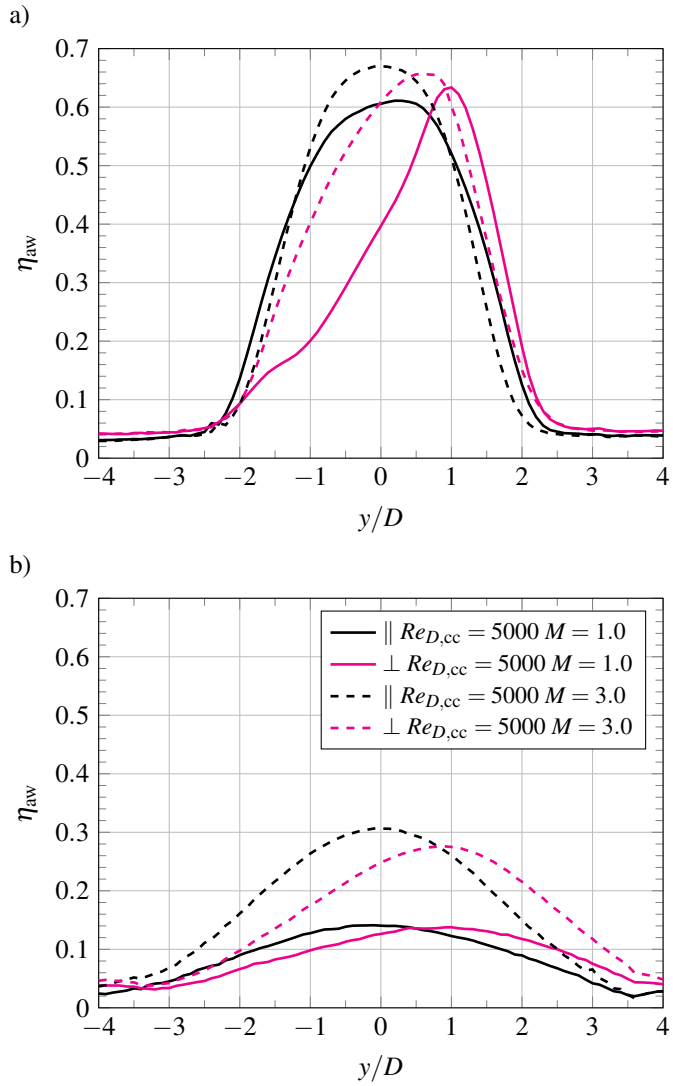

FIGURE 7: Adiabatic film cooling effectiveness η_{aw} at $M = 1.0$ and $M = 3.0$ for parallel ($\|$) and perpendicular ($\perp$) coolant flow ($Re_{D,\text{cc}} = 5000$) against y/D at $x/D = 5$ (a) and $x/D = 40$ (b) (Legend applicable for both graphs)

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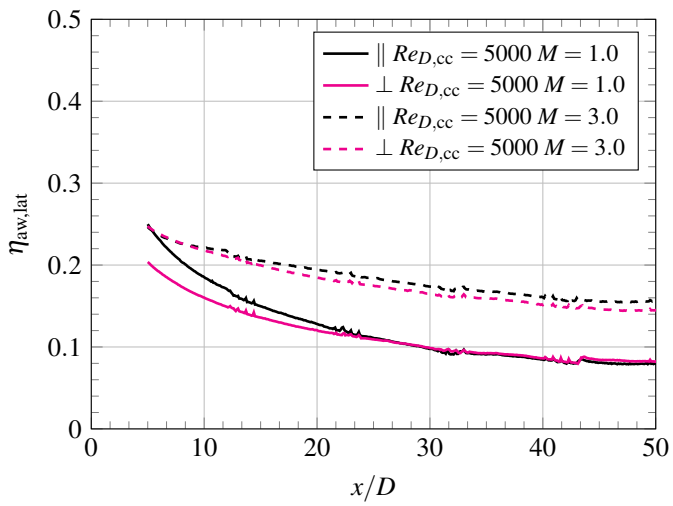

FIGURE 8: Laterally averaged adiabatic film cooling effectiveness $\eta_{\mathrm{aw,lat}}$ for parallel coolant flow ($\parallel$) and perpendicular coolant flow ($\perp$) at blowing ratios $M = 1.0$ and $M = 3.0$ and $5 \leq x/D \leq 50$

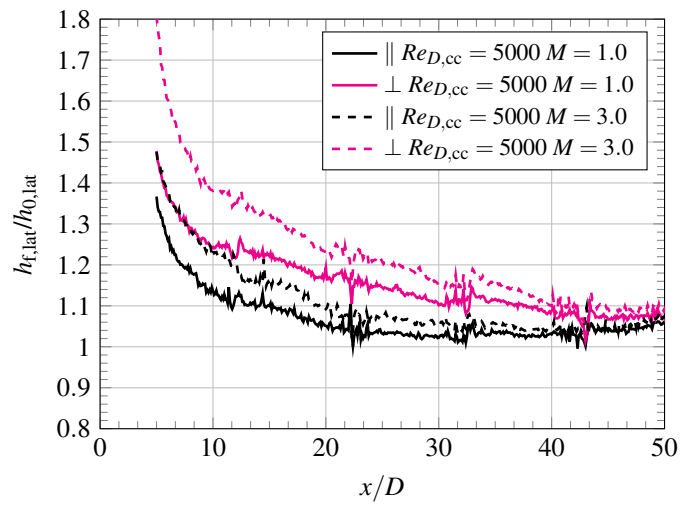

FIGURE 9: Laterally averaged heat transfer coefficients $h_{\mathrm{f,lat}}/h_{0,\mathrm{lat}}$ for parallel coolant flow ($\parallel$) and perpendicular coolant flow ($\perp$) at blowing ratios $M = 1.0$ and $M = 3.0$ and $5 \leq x/D \leq 50$

Figure 8 shows laterally averaged values of the adiabatic film cooling effectiveness for both cooling channel orientations and the above-mentioned blowing ratios in streamwise direction in order to evaluate the global effect of coolant flow orientation on the performance of the cooling hole. The results for the parallel coolant flow have already been discussed in Fig. 4. At the higher blowing ratio $M = 3.0$, the laterally averaged film cooling effectiveness is just slightly reduced compared to parallel coolant flow orientation. This could already be deduced from the respective contour plots in Figs. 5a and b and is, hence, not unexpected. Figure 5b reveals that the coolant jet is shifted towards the opposite direction of the coolant flow. However, the coverage as well as the decay in film cooling effectiveness are quite similar for both coolant flow orientations. This is not the case for the lowest blowing ratio $M = 1.0$. In this case, the contraction of the coolant jet is much stronger for the perpendicular coolant flow direction than for the parallel one. It is most distinct in the region close to the cooling hole exit for $x/D \leq 20$. Thus, the laterally averaged film cooling effectiveness is considerably lower for perpendicular coolant flow direction. In the region further downstream, the lines approach each other and merge at $x/D \approx 24$.

Film cooling is not only characterized by film cooling effectiveness, but also by heat transfer variation. Laterally averaged heat transfer coefficients $h_{\mathrm{f,lat}}$ are shown in Fig. 9. They are normalized by the heat transfer coefficients without film cooling $h_{0,\mathrm{lat}}$. The values exceed one for all cases with a few exceptions. This means that heat transfer is augmented due to film cooling. Close to the cooling hole exit, heat transfer augmentation is highest for all cases. It then decreases with a steadily decreasing slope until the end of the measurement area at $x/D = 50$. Minor perturbations are visible in the overlapping area of the sapphire windows. At both coolant flow orientations heat transfer is increased at a higher blowing ratio. This is due to higher coolant jet velocities at higher blowing ratios. Such an effect of the blowing ratio on the heat transfer was also reported by Saumweber et al. [15]. At the end of the measurement area, no effect of blowing ratio is present. Heat transfer is still increased for all cases. The change in coolant flow orientation from parallel to perpendicular leads to a significant increase in heat transfer by more than 10 percentage points for $x/D < 25$. Highest values for the ratio of heat transfer coefficients can be observed for the highest blowing ratio $M = 3.0$ and perpendicular coolant flow orientation. In this case, the increase in heat transfer for the region close to the cooling hole ($x/D \leq 10$) is more than 40 %.

An increase in heat transfer due to perpendicular coolant flow orientation is new. Saumweber and Schulz [15] reported a lower augmentation compared to parallel coolant flow orientation. In their study, however, the perpendicular coolant flow effect was stronger due to high coolant flow velocities and the contact area between surface and jet was reduced largely. Additionally, the measurement area extended only up to $x/D = 10$ and the cooling hole shape was different.

For perpendicular coolant flow, it is assumed that the turbulence of the film cooling jet is higher due to the separation and strong deflection at cooling hole inlet. A higher turbulence is known to increase heat transfer and thus believed to be the main reason for the higher heat transfer for perpendicular coolant flow orientation. To further investigate the effect of increased

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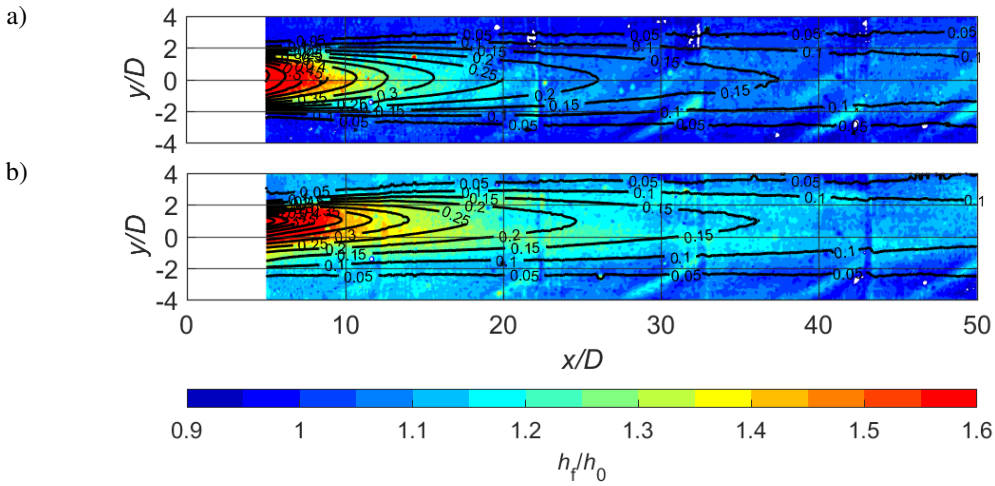

FIGURE 10: Contours of heat transfer coefficients $h_{\mathrm{f,lat}}/h_{0,\mathrm{lat}}$ for $5 \leq x/D \leq 50$ and $-4 \leq y/D \leq 4$ at $M = 1.0$ and $Re_{D,\mathrm{cc}} = 5000$ with lines of constant adiabatic film cooling effectiveness (black)
a) Parallel coolant flow orientation ($\|$) b) Perpendicular coolant flow orientation ($\perp$)

heat transfer, local heat transfer augmentation and lines of constant adiabatic film cooling effectiveness are shown in Fig. 10. Figure 10a shows the parallel coolant flow orientation, whereas Fig. 10b shows the case of perpendicular coolant flow. Even though the lines of constant adiabatic film cooling effectiveness are shifted towards the positive y-direction, heat transfer augmentation is not influenced in the same way. Close to the cooling hole exit, heat transfer augmentation appears to be shifted as well. However, further downstream at $x/D > 15$, the location of highest heat transfer augmentation is found to be near the centerline. A similar effect was discussed by Anderson et al. [32] for compound angle shaped holes, where heat transfer was increased in areas between the cooling holes where no coolant jet was present. It was assumed the a vortex developes when the free stream passes over the inclined jet. This vortex rotates back towards the wall and increases heat transfer in areas where no coolant jet is present. Since the jet in the perpendicular coolant flow orientation is inclined as well, it is assumed that a similar effect takes place, leading to an increased heat transfer close to the centerline. In addition, heat transfer is also increased in the areas where the jet is present. Consequently, heat transfer is increased in a wide area. The higher heat transfer for perpendicular coolant flow is thus believed to be due to the combined effect of a wider area with increased heat transfer and a higher turbulence inside the coolant jet.

Effect of Coolant Reynolds Number

In this section, the cooling channel Reynolds number is varied from $Re_{D,\mathrm{cc}} = 5000$ to $Re_{D,\mathrm{cc}} = 30000$ in order to investigate its influence on film cooling. Results are shown for perpendicular coolant flow orientation only, as the effect is expected to be more distinct for this setup. Contours of the adiabatic film cooling effectiveness for the higher coolant channel Reynolds number are shown in Fig. 5c. Values for the adiabatic film cooling effectiveness are lower at all blowing ratios compared to the other cases (Figs. 5a and b) and a shift of the coolant jet is visible. The shift is oriented towards the positive y-direction opposite to the lower coolant Reynolds number (Fig. 5b). Such a change in the shift was reported by others as well (see e.g. McClintic et al. [19]). It can be explained by the combination of flow separation at the cooling hole inlet and the helical flow inside the hole. The wake of the separation bubble rotates inside the cylindrical part of the coolant hole and the flow will separate at the diffuser inlet at the position of the wake. A change in velocity ratio $u_{\mathrm{cc}}/u_{\mathrm{c}}$ changes the helical motion and consequently the position of the diffuser separation.

Figures 11a and b show the distribution of adiabatic film cooling effectiveness for the blowing ratios $M = 1.0$ and 3.0 at $x/D = 5$ and $x/D = 40$, respectively. At the higher coolant channel Reynolds number, values for the adiabatic film cooling effectiveness are much lower for both blowing ratios compared to the lower coolant channel Reynolds number. At the position close to the cooling hole exit at $x/D = 5$, the maximum effectiveness for $Re_{D,\mathrm{cc}} = 30000$ and $M = 3.0$ is about $\eta_{\mathrm{aw}} \approx 0.48$. This is somewhat higher than that of the lower blowing ratio, which amounts to $\eta_{\mathrm{aw}} \approx 0.44$. The spreading of the coolant on the measurement surface is lower at the higher blowing ratio. For the higher Reynolds number, the location of the maximum adiabatic film cooling effectiveness is shifted towards the negative y-direction and is located at approximately $y/D = -1.2$ and $y/D = -0.7$ for $M = 1.0$ and $M = 3.0$, respectively. The shift

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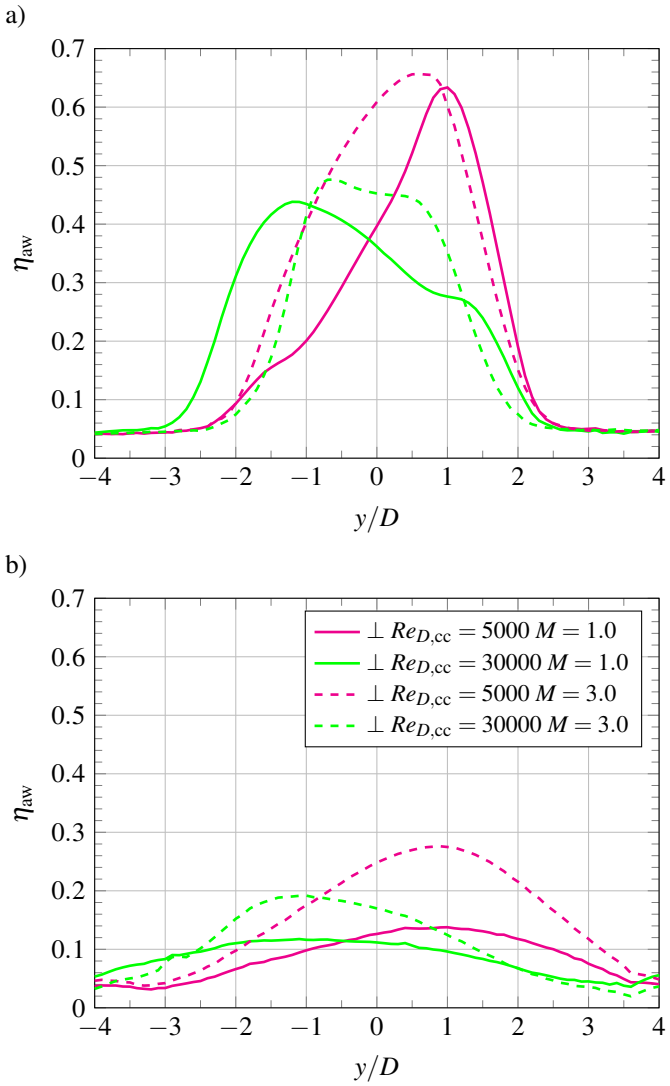

FIGURE 11: Adiabatic film cooling effectiveness η_{aw} at $M = 1.0$ and $M = 3.0$ for $Re_{D,cc} = 5000$ and $Re_{D,cc} = 30000$ (perpendicular coolant flow ($\perp$)) against y/D at $x/D = 5$ (a) and $x/D = 40$ (b) (Legend applicable for both graphs)

of the maximum is directed oppositely to the case of the lower Reynolds number. At the position near the downstream end of the measurement area at $x/D = 40$ (see Fig. 11b), values of the adiabatic film cooling effectiveness are also lower at the higher coolant Reynolds number for both blowing ratios. As mentioned earlier, this is attributed to the fact that flow separation occurs in the diffuser part of the cooling holes. The flow separation shifts to the opposite side for higher coolant crossflow and changes the mixing of the coolant with the hot gas. The maximum adiabatic film cooling effectiveness shifts even further towards the negative y-direction at the higher blowing ratio $M = 3.0$. This is not

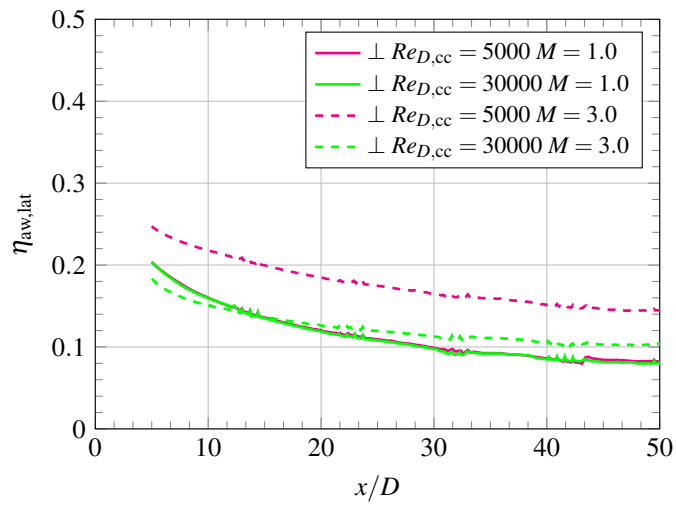

FIGURE 12: Laterally averaged adiabatic film cooling effectiveness $\eta_{aw,lat}$ against $5 \leq x/D \leq 50$ for perpendicular coolant flow ($\perp$) and Reynolds numbers $Re_{D,cc} = 5000$ and 30000 at blowing ratios $M = 1.0$ and $M = 3.0$

the case for the lower blowing ratio $M = 1.0$. Here, the position of the maximum remains unchanged compared to the area close to the cooling hole exit shown in Fig. 11a.

Figure 12 shows the laterally averaged adiabatic film cooling effectiveness. This presentation helps to capture the overall effect of coolant Reynolds number on cooling effectiveness. Compared to the lower coolant Reynolds number $Re_{D,cc} = 5000$, values for the higher blowing ratio $M = 3.0$ are clearly lower for $Re_{D,cc} = 30000$. For the lower blowing ratio $M = 1.0$, the results for both Reynolds numbers are in good qualitative and quantitative agreement. Compared to the other cases, laterally averaged adiabatic film cooling effectiveness values are the lowest at the higher blowing ratio $M = 3.0$ and coolant Reynolds number in the region close to the cooling hole exit. Towards the downstream end of the measurement plate, however, the values for $M = 3.0$ and $Re_{D,cc} = 30000$ are higher than those for the lower blowing ratios at both Reynolds numbers. Compared to the lower blowing ratio cases, the amount of coolant introduced into the hot gas flow is far higher, but the severe flow separation in the diffuser and the resulting mixing of the flow lead to a low cooling effectiveness. Therefore, blowing ratio has little effect on cooling effectiveness for high coolant channel Reynolds numbers and perpendicular coolant flow orientation. The trends discussed are in good agreement with Saumweber and Schulz [15]. They investigated a case with about the same velocity ratios u_{cc}/u_c as those in the current higher coolant Reynolds number case. Results showed a deep drop of cooling effectiveness at high blowing ratios for a perpendicular coolant flow. Values for adiabatic ef-

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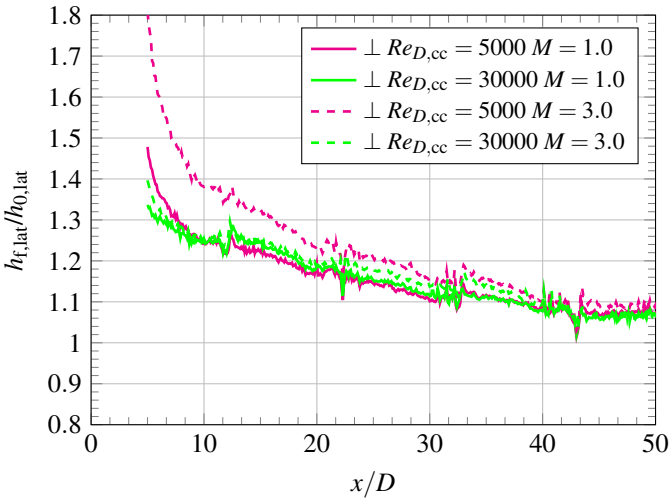

FIGURE 13: Ratio of heat transfer coefficients $h_{f,lat}/h_0$ for perpendicular coolant flow against $5 \leq x/D \leq 50$ at blowing ratios $M = 1.0$ and $M = 3.0$ and Reynolds numbers $Re_{D,cc} = 5000$ and 30000

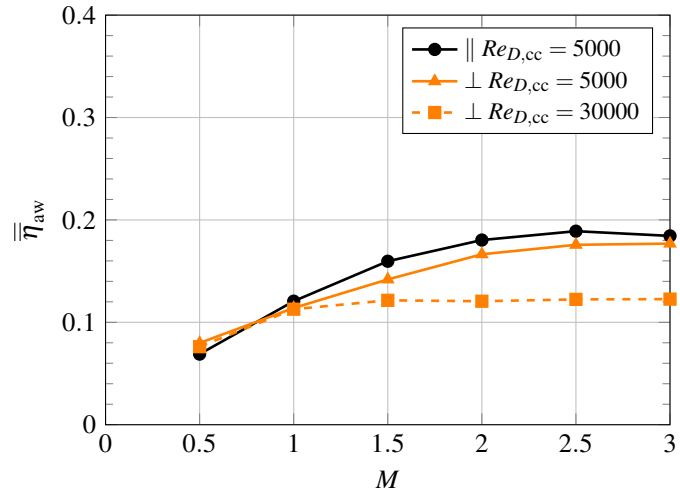

FIGURE 14: Spatially averaged adiabatic film cooling effectiveness $\overline{\overline{\eta}}_{aw}$ for blowing ratios $0.5 \leq M \leq 3.0$

fectiveness were lower in the near-hole region than in cases with the same coolant flow velocity, but lower blowing ratios.

In order to investigate the effect of coolant Reynolds number on film cooling, the change in heat transfer due to film cooling has to be examined. The ratio of laterally averaged heat transfer coefficients $h_{f,lat}/h_{0,lat}$ for blowing ratios $M = 1.0$ and $M = 3.0$ and Reynolds numbers $Re_{D,cc} = 5000$ and 30000 is shown in Fig. 13. All test cases shown in this diagram reveal an increase of heat transfer due to film cooling ($h_{f,lat}/h_{0,lat} > 1$). Generally, heat transfer augmentation decreases with surface length and is lower than 20% for $x/D > 20$. With increasing length, all lines converge. The effect of blowing ratio on heat transfer is minor for the high coolant channel Reynolds number. Consequently, blowing ratio effects on heat transfer and film cooling effectiveness are minor at high coolant channel Reynolds numbers. At the low blowing ratio $M = 1.0$ and $Re_{D,cc} = 5000$, heat transfer augmentation is approximately the same as at both blowing ratios for the higher coolant Reynolds number $Re_{D,cc} = 30000$ and $x/D > 10$. In the near hole region, however, heat transfer is the lowest for the higher coolant Reynolds number $Re_{D,cc} = 30000$. Values are even lower than for the parallel coolant flow orientation, see Fig. 9. This is in agreement with the results of Saumweber et al. [15] who reported a lower heat transfer for perpendicular coolant flow orientation and $x/D < 10$ due to a reduced contact between coolant jet and surface at high velocity ratios u_{cc}/u_c. The $Re_{D,cc} = 5000$, $M = 3.0$ case on the other hand shows the highest heat transfer augmentation. For this case, the contact area between coolant jet and surface is barely reduced and turbulence inside the coolant jet is believed to be higher than for the parallel coolant flow orientation due to a stronger flow separation at the cooling hole inlet. The combination of both effects lead to high heat transfer in the near hole region. Further downstream, all lines converge. In general, heat transfer is increased due to perpendicular coolant flow orientation in most areas.

Spatially averaged adiabatic film cooling effectiveness is evaluated for all investigated test cases at blowing ratios $0.5 \leq M \leq 3.0$ and is shown in Fig. 14. The results match the results previously shown in this study and give a good summary of all effects. For all cases, the values of the spatially averaged adiabatic film cooling effectiveness increase with increasing blowing ratio until they reach a plateau. The performance of the film cooling hole investigated is best for the parallel coolant flow configuration. The curve for the perpendicular coolant flow orientation and the lower coolant Reynolds number is similar to that of parallel coolant flow. The curves start with approximately the same value for the lowest blowing ratio and are separated by a constant offset of $\Delta\overline{\overline{\eta}}_{aw} \approx 0.01$ for $M \geq 1$. Cooling performance is worst for the higher coolant Reynolds number. A plateau is reached at $M = 1.5$ and an increase of added coolant does not lead to any increase in spatially averaged cooling effectiveness. This clearly shows the drawback of the laidback fan-shaped hole investigated in this study. Since perpendicular coolant flow is very common in modern gas turbine blades, there is room for improvement and more studies will be carried out in the near future in order to optimize the shape of the cooling holes for perpendicular coolant flow.

CONCLUSIONS

Film cooling measurements for a 10-10-10 laidback fan-shaped hole with coolant crossflow were presented and discussed

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in this paper. Data were evaluated up to $50D$ downstream of the cooling hole exit. The effects of coolant flow orientation as well as of the coolant channel Reynolds number on adiabatic film cooling effectiveness and heat transfer variation were analyzed.

For a parallel coolant flow, film cooling effectiveness increases with increasing blowing ratio. At blowing ratios $M > 2.5$, the jet contracts and effectiveness is slightly reduced. Changing the channel orientation from parallel to perpendicular coolant flow has a severe impact on both cooling effectiveness and heat transfer. Film cooling effectiveness is reduced by about 6 % at a blowing ratio of $M = 1.0$. An asymmetric separation in the diffuser of the shaped hole leads to an asymmetric footprint of the coolant jet on the surface with reduced effectiveness. The effect is more pronounced for a low blowing ratio.

For a higher coolant channel Reynolds number ($Re_{D,\mathrm{cc}} = 30000$) and perpendicular coolant flow, asymmetry increases and film cooling effectiveness decreases. There are only minor effects of blowing ratio on film cooling effectiveness and heat transfer. An increasing blowing ratio does not lead to a significant increase in effectiveness as it is the case for low coolant channel Reynolds numbers. Thus, a high coolant channel Reynolds number results in a drop in film cooling effectiveness especially for higher blowing ratios. Compared to the lower coolant channel Reynolds number case, there is a major difference in film cooling effectiveness of 30 % at $M = 3.0$. The maximum film cooling effectiveness is shifted from one side of the centerline to the other.

Heat transfer is increased due to film cooling for both parallel and perpendicular coolant flow. In the near hole region, increase is higher than 30 %. Downstream of the cooling hole exit, heat transfer augmentation decreases. For all perpendicular coolant flow cases and $10 < x/D < 25$, heat transfer is more than 10 percentage points higher than for the parallel coolant flow orientation. In the near hole region, heat transfer is the lowest for the high coolant channel Reynolds number, since contact area between surface and coolant jet is reduced significantly in this case.

Results prove that coolant flow orientation has a severe impact on both film cooling effectiveness and heat transfer. New cooling hole shapes will be investigated in near-future studies aiming to improve film cooling effectiveness distribution in perpendicular coolant flow cases.

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References

[1] Schroeder, R. P., and Thole, K. A., 2014. "Adiabatic Effectiveness Measurements for a Baseline Shaped Film Cooling Hole". In Proceedings of ASME Turbo Expo 2014, No. GT2014-25992.

[2] Saumweber, C., Schulz, A., and Wittig, S., 2003. "Free-Stream Turbulence Effects on Film Cooling With Shaped Holes". *Journal of Turbomachinery,* ***125***(1), pp. 65–73.

[3] Anderson, J. B., Wilkes, E. K., McClintic, J. W., and Bogard, D. G., 2016. "Effects of Freestream Mach Number, Reynolds Number, and Boundary Layer Thickness on Film Cooling Effectiveness of Shaped Holes". In Proceedings of ASME Turbo Expo 2016, No. GT2016-56152.

[4] Bunker, R. S., 2005. "A Review of Shaped Hole Turbine Film-Cooling Technology". *Journal of Heat Transfer,* ***127***(4), pp. 441–453.

[5] Schmidt, D. L., Sen, B., and Bogard, D. G., 1996. "Effects of Surface Roughness on Film Cooling". In Proceedings of ASME Turbo Expo 1996, No. 96-GT-299.

[6] Bogard, D. G., and Thole, K. A., 2006. "Gas Turbine Film Cooling". *Journal of Propulsion and Power,* ***22***(2), pp. 249–270.

[7] Fraas, M., Glasenapp, T., Schulz, A., and Bauer, H.-J., 2017. "Introducing a New Test Rig for Film Cooling Measurements with Realistic Hole Inflow Conditions". In Proceedings of ASME Turbo Expo 2017, No. GT2017-63585.

[8] Leylek, J. H., and Zerkle, R. D., 1994. "Discrete-Jet Film Cooling: A Comparison of Computational Results With Experiments". *Journal of Turbomachinery,* ***116***(3), pp. 358–368.

[9] Acharya, S., and Houston Leedom, D., 2013. "Large Eddy Simulations of Discrete Hole Film Cooling With Plenum Inflow Orientation Effects". *Journal of Heat Transfer,* ***135***(1), p. 011010.

[10] Thole, K. A., Gritsch, M., Schulz, A., and Wittig, S., 1997. "Effect of a Crossflow at the Entrance to a Film-Cooling Hole". *Journal of Fluids Engineering,* ***119***(3), pp. 533–540.

[11] Gritsch, M., Schulz, A., and Wittig, S., 1998. "Adiabatic Wall Effectiveness Measurements of Film-Cooling Holes With Expanded Exits". *Journal of Turbomachinery,* ***120***(3), pp. 549–556.

[12] Saumweber, C., Schulz, A., Wittig, S., and Gritsch, M., 2001. "Effects of Entrance Crossflow Directions to Film Cooling Holes". *Annals of the New York Academy of Sciences,* ***934***(1), pp. 401–408.

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[13] Gritsch, M., Schulz, A., and Wittig, S., 1998. "Heat Transfer Coefficient Measurements of Film-Cooling Holes With Expanded Exits". In Proceedings of ASME Turbo Expo 1998, No. 98-GT-028.

[14] Kohli, A., and Thole, K. A., 1998. "Entrance Effects on Diffused Film-Cooling Holes". In Proceedings of ASME Turbo Expo 1998, No. 98-GT-402.

[15] Saumweber, C., and Schulz, A., 2008. "Comparison the Cooling Performance of Cylindrical and Fan-Shaped Cooling Holes With Special Emphasis on the Effect of Internal Coolant Cross-Flow". In Proceedings of ASME Turbo Expo 2008, No. GT2008-51036.

[16] Gritsch, M., Schulz, A., and Wittig, S., 2003. "Effect of Internal Coolant Crossflow on the Effectiveness of Shaped Film-Cooling Holes". *Journal of Turbomachinery,* ***125***(3), pp. 547–554.

[17] McClintic, J. W., Klavetter, S. R., Winka, J. R., Anderson, J. B., Bogard, D. G., Dees, J. E., Laskowski, G. M., and Briggs, R., 2015. "The Effect of Internal Crossflow on the Adiabatic Effectiveness of Compound Angle Film Cooling Holes". *Journal of Turbomachinery,* ***137***(7), p. 071006.

[18] Wilkes, E., Anderson, J., McClintic, J., and Bogard, D., 2016. "An Investigation of Turbine Film Cooling Effectiveness With Shaped Holes and Internal Cross-Flow With Varying Operational Parameters". In Proceedings of ASME Turbo Expo 2016, No. GT2016-56162.

[19] McClintic, J. W., Anderson, J. B., Bogard, D. G., Dyson, T. E., and Webster, Z. D., 2018. "Effect of Internal Crossflow Velocity on Film Cooling Effectiveness—Part I: Axial Shaped Holes". *Journal of Turbomachinery,* ***140***(1), p. 011003.

[20] Dittmar, J., Schulz, A., and Wittig, S., 2003. "Assessment of Various Film-Cooling Configurations Including Shaped and Compound Angle Holes Based on Large-Scale Experiments". *Journal of Turbomachinery,* ***125***(1), pp. 57–64.

[21] McClintic, J. W., Anderson, J. B., Bogard, D. G., Dyson, T. E., and Webster, Z. D., 2018. "Effect of Internal Crossflow Velocity on Film Cooling Effectiveness—Part II: Compound Angle Shaped Holes". *Journal of Turbomachinery,* ***140***(1), p. 011004.

[22] Sakai, E., and Takahashi, T., 2011. "Experimental and Numerical Study on Effects of Turbulence Promoters on Flat Plate Film Cooling". In Proceedings of ASME Turbo Expo 2011, No. GT2011-45196.

[23] Choe, H., Kays, W. M., and Moffat, R. J., 1974. "The Superposition Approach to Film-Cooling". In Proceedings of ASME Winter Annual Meeting 1974.

[24] Gritsch, M., Baldauf, S., Martiny, M., Schulz, A., and Wittig, S., 1999. "The Superposition Approach to Local Heat Transfer Coefficients in High Density Ratio Film Cooling Flows". In Proceedings of ASME Turbo Expo 1999, No. 99-GT-168.

[25] Kneer, J., Puetz, F., Schulz, A., and Bauer, H.-J., 2014. "Application of the Superposition Principle of Film-Cooling on a Non-Axisymmetric Turbine Endwall". In 15th International Symposium on Transport Phenomena and Dynamics of Rotating Machinery.

[26] Kneer, J., Puetz, F., Schulz, A., and Bauer, H.-J., 2016. "A New Test Facility to Investigate Film Cooling on a Non-axisymmetric Contoured Turbine Endwall—Part II: Heat Transfer and Film Cooling Measurements". *Journal of Turbomachinery,* ***138***(7), p. 071004.

[27] Ochs, M., Schulz, A., and Bauer, H.-J., 2010. "High Dynamic Range Infrared Thermography by Pixelwise Radiometric Self Calibration". *Infrared Physics & Technology,* ***53***(2), pp. 112–119.

[28] Ochs, M., Horbach, T., Schulz, A., Koch, R., and Bauer, H.-J., 2009. "A Novel Calibration Method for an Infrared Thermography System Applied to Heat Transfer Experiments". *Measurement Science and Technology,* ***20***(7), p. 075103.

[29] Kline, S. J., and McClintock, F. A., 1953. "Describing Uncertainties in Single-Sample Experiments". *Mechanical Engineering,* ***75***(1), pp. 3–8.

[30] Saumweber, C., and Schulz, A., 2012. "Effect of Geometry Variations on the Cooling Performance of Fan-Shaped Cooling Holes". *Journal of Turbomachinery,* ***134***(6), p. 061008.

[31] Haydt, S., Lynch, S., and Lewis, S., 2017. "The Effect of Area Ratio Change via Increased Hole Length for Shaped Film Cooling Holes with Constant Expansion Angles". In Proceedings of ASME Turbo Expo 2017, No. GT2017-63692.

[32] Anderson, J. B., McClintic, J. W., Bogard, D. G., Dyson, T. E., and Webster, Z. D., 2017. "Freestream Flow Effects on Film Effectiveness and Heat Transfer Coefficient Augmentation for Compound Angle Shaped Holes". In Proceedings of ASME Turbo Expo 2017, No. GT2017-64853.

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